Prasita Mukherjee

Ph.D. student, Purdue University

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https://github.com/prasitaGit

Research Interests

My research interests lie in the areas of neurosymbolic and automated reasoning. My research focuses on broadening the scope of hard problems in the domain of formal methods and programming languages by providing learning-based solutions.

Education

Ph.D. Computer Science *Purdue University* **Indiana, USA** *2019 - present*
Advisor: Prof. Benjamin Delaware

M.Tech Computer Science *Indian Institute Of Technology, Guwahati* **Assam, India** *2017 - 2019*
Thesis Advisor: Prof. Purandar Bhaduri (Thesis)

B.Sc + M.Sc. Computer Science *University Of Calcutta* **West Bengal, India** *2011 - 2016*

Professional Experience

Research Intern *(Bloomberg)* **New York, USA** *05/2025 - 08/2025*
- Differential Random Testing of a Compiler (Mentor: Joomy Korkut)

Graduate Teaching Assistant *(Purdue University)* **Indiana, USA** *08/2019 - present*
- Courses: CS 251 (Data Structures and Algorithms), CS 252 (Systems Programming), CS 456 (Programming Languages)

Applied Science Intern *(Amazon)* **Washington, USA** *05/2022 - 08/2022*
- Developed initial SMT encoding for the authorization system CEDAR (Manager: Emina Torlak)

Teaching Assistant *Indian Institute Of Technology, Guwahati* **Assam, India** *08/2017 - 05/2019*
- Courses: Introduction to C, Data Structures, Formal Methods

Software Engineer *(Polaris Networks, Inc.)* **West Bengal, India** *07/2016 - 06/2017*
- Front end engineer for Evolved Packet Core (EPC) in 4G-LTE network

Publications

P. Mukherjee, R. Sowards, J. DiVincenzo, B. Delaware. **SMEAGOL**: Automated Inference and Testing of Separation Logic Specifications with Magic Wands (**Under Submission**) [draft]

P. Mukherjee, M. Lu, B. Delaware. **SYNVER**: Towards Automated Verification of LLM-Synthesized C Programs (**ASE 2025** - Full Paper, **CoqPL 2025** - Extended abstract) [paper]

R. Dickerson, **P. Mukherjee**, B. Delaware. **KestREL**: Relational Verification using E-Graphs for Program Alignment (**OOPSLA 2025**) [paper]

P. Mukherjee and H. Yin. OCTAL: Graph Representation Learning for LTL Model Checking. **DLG-KDD 2023** - Selected as a Contributed Talk [paper]

K. Nagar, **P. Mukherjee** and S. Jagannathan. Semantics, Specification, and Bounded Verification of Concurrent Libraries in Replicated Systems. **CAV 2020**.[paper]

S. Banerjee, **P. Mukherjee**, S. Kanrar, and N. Chaki. A Novel Symmetric Algorithm for Process Synchronization in Distributed Systems. **ALAP 2018**. [paper]

Awards and Travel Grants

- One-time employee recognition award (Purdue University, Amount - 1500 USD)
- Travel grant for KDD 2023 (Purdue University, Amount - 500 USD)